

Part 1. Text problems. In section 3.4 (pdf version of the text) do problems 196, 198, 203, 205, 206. Note that 206 needs to be factored a lot, and this section can be time consuming but I'll show us some tricks to try to dodge some of worst of it if needed.

Part 2. Mixed Practice. Expect to use integration by parts, algebra tricks, u -subs, the table of integrals on page 617 and anything else available. I expect questions on this throughout the week(end). Please be sure to look at these/make rough attempts before then.

1) Find the following integral.

$$\int \frac{x-1}{x^2+9x-10} dx$$

2) Find the following integral.

$$\int \frac{1}{x^2+4x+4} dx$$

3) Find the following integral.

$$\int x \sec^2(x) dx$$

4) Find the following integral.

$$\int \frac{x-3}{x^2+1} dx$$

5) Find the following integral.

$$\int \frac{1}{x^2+6x+25} dx$$

6) Find the following integral.

$$\int \frac{1+e^x}{1-e^x} dx$$

Part 3. From the “Forbidden Textbook” please choose three problems (other than question 19) that send an initial shiver down your spine. For each problem I want you to do a pre- and post- analysis. Before attempting the problem write a sentence or two explaining why this particular integral looked challenging to you. This can range from reasons including ‘I’m not comfortable with trigonometry’ or ‘there’s no obvious first move’ etc. Then perform each problem and write a sentence identifying if the process seems more clear with reasons. I’ve provided an example response for question 19 on the next page.

‘Given the problem $\int e^{x+e^x} dx$ has a very awkward exponent, I’m unsure if there is an obvious substitution or some algebraic greasing-of-the-wheels that must be done first. My initial impression is that u should equal the junk in the exponent, but that might not pan out. Let me try some algebra and see if it becomes more clear.

$$\int e^{x+e^x} dx = \int e^x \cdot e^{e^x} dx =$$

Aha, by rewriting it I can take a better guess at a non-obvious substitution of $u = e^x$ and see where that leads.

Let $u = e^x$, then $du = e^x dx$ and our integral becomes

$$\int e^x e^{e^x} dx = \int e^u du = e^u + C = e^{e^x} + C$$

In hindsight, the jumbling together of the exponents using exponent laws hid the substitution from me, by untangling that it became a bit more obvious what a good first attempt a u -sub would be and that happened to work. Huzzah!’

Your answer need not look exactly the same and your reasoning should be wholly your own. I am convinced this was a past homework question or cannon fodder for the notes hence its exclusion from the list. Also, don’t pick any hyperbolic trig stuff as we simply haven’t covered it.

part 1

$$196 \int \frac{dx}{(x-3)(x-2)}$$

$$\frac{1}{(x-3)(x-2)} = \frac{A}{x-3} + \frac{B}{x-2} \quad \begin{array}{l} 1 = A(x-2) + B(x-3) \\ A=1 \quad B=-1 \end{array}$$

$$\int \left(\frac{1}{x-3} - \frac{1}{x-2} \right) dx = \ln|x-3| - \ln|x-2| + C$$
$$\ln \left| \frac{x-3}{x-2} \right| + C$$

$$198 \int \frac{dx}{x^3-x}$$

$$\int \frac{dx}{x(x^2-1)}$$

$$x(x-1)(x+1)$$

$$\frac{1}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{(x-1)} + \frac{C}{(x+1)}$$

$$1 = A(x-1)(x+1) + Bx(x-1) + Cx(x+1)$$

$$A=-1 \quad B=\frac{1}{2} \quad C=\frac{1}{2}$$

$$\int \left(-\frac{1}{x} + \frac{1}{2(x-1)} + \frac{1}{2(x+1)} \right) dx$$

$$-\ln|x| + \frac{1}{2}\ln|x-1| + \frac{1}{2}\ln|x+1| + C$$

$$-\ln|x| + \frac{1}{2}\ln|x^2-1| + C$$

$$203 \int \frac{2-x}{x^2+2} dx$$



$$\int \frac{x}{x^2+2} dx$$

$$u = x^2 + 2$$

$$du = 2x dx \Rightarrow x dx = \frac{1}{2} du$$

$$\int \frac{\frac{1}{2} du}{u}$$

$$\frac{1}{2} \int \frac{1}{u} du$$

$$\frac{1}{2} \ln |u|$$

$$\frac{1}{2} \ln |x^2 + 2|$$

$$\int \frac{2}{x^2+2} dx - \int \frac{x}{x^2+2} dx$$

$$2 \int \frac{1}{x^2 + \sqrt{2}^2} dx$$

$$2 \cdot \frac{1}{\sqrt{2}} \arctan\left(\frac{x}{\sqrt{2}}\right)$$

$$\sqrt{2} \arctan\left(\frac{x}{\sqrt{2}}\right)$$

$$\int \frac{2-x}{x^2+2} dx = \sqrt{2} \arctan\left(\frac{x}{\sqrt{2}}\right) - \frac{1}{2} \ln |x^2 + 2| + C$$

$$205 \int \frac{dx}{x^3 - 2x^2 - 4x + 8}$$

$$x^3 - 2x^2 - 4x + 8$$

$$x^2(x-2) - 4(x-2)$$

$$(x^2 - 4)(x-2)$$

$$(x+2)(x-2)^2$$

$$\frac{1}{(x+2)(x-2)(x-2)} = \frac{A}{x+2} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$$

$$1 = A(x-2)^2 + B(x+2)(x-2) + C(x+2)$$

$$A = \frac{1}{16}$$

$$B = -\frac{1}{16}$$

$$C = \frac{1}{4}$$

$$\int \left(\frac{1}{16(x+2)} - \frac{1}{16(x-2)} + \frac{1}{4(x-2)^2} \right) dx$$

$$\frac{1}{16} \ln \left| \frac{x+2}{x-2} \right| - \frac{1}{4(x-2)} + C$$

$$206 \int \frac{dx}{x^4 - 10x^2 + 9}$$

* needs to be factored a lot

Part 2

$$1) \int \frac{x-1}{x^2+9x-10} dx$$

$$\int \frac{\cancel{x-1}}{(x+10)(\cancel{x-1})} dx$$

$$\frac{1}{x+10}$$

$$\ln |x+10| + C$$

$$2) \int \frac{1}{x^2+4x+4} dx$$

$$\int \frac{1}{(x+2)^2} dx$$

$$\int (x+2)^{-2} dx$$

$$\int u^{-2} du$$

$$\frac{u^{-2+1}}{-2+1} + C$$

$$\frac{u^{-1}}{-1} + C$$

$$-\frac{1}{u} + C$$

$$-\frac{1}{x+2} + C$$

$$3) \int x \sec^2(x) dx \quad \int u dv = uv - \int v du$$

$$u = x \quad du = dx$$

$$dv = \sec^2 x \quad v = \tan x$$

$$\int x \sec^2 x dx = x \cdot \tan x - \int \tan x dx$$

$$\int \tan x dx = \int \frac{\sin(x)}{\cos(x)} dx$$

$$w = \cos x$$

$$dw = -\sin x dx$$

$$\int \frac{-dw}{w}$$

$$-\ln|w|$$

$$-\ln|\cos x|$$

$$x \tan x + \ln|\cos x| + C$$

$$4) \int \frac{x-3}{x^2+1} dx$$

$$\int \frac{x}{x^2+1} dx - \frac{3}{x^2+1}$$

$$u = x^2 + 1$$

$$du = 2x dx \rightarrow x dx \rightarrow \frac{1}{2} du$$

$$\frac{1}{2} \int \frac{1}{u} du$$

$$\frac{1}{2} \ln |u|$$

$$\frac{1}{2} \ln(x^2 + 1)$$

$$\frac{1}{2} \ln(x^2 + 1) - 3 \arctan x + C$$

$$\frac{3}{x^2 + 1}$$

$$3 \int \frac{1}{x^2 + 1} dx$$

$$3 \arctan x$$

$$5) \int \frac{1}{x^2 + 6x + 25} dx$$

$$(x^2 + 6x + 9) + 16$$

$$(x+3)^2 + 4^2$$

$$\int \frac{1}{(x+3)^2 + 4^2} dx$$

$$u = x+3$$

$$du = dx$$

$$\int \frac{1}{u^2 + 4^2} du$$

$$\frac{1}{4} \arctan \frac{u}{4} + C$$

$$\frac{1}{4} \arctan \left(\frac{x+3}{4} \right) + C$$

$$6) \int \frac{1+e^x}{1-e^x} dx$$

$$u = e^x$$

$$du = e^x dx \rightarrow u dx \rightarrow dx \rightarrow \frac{du}{u}$$

$$\int \frac{1+u}{1-u} \cdot \frac{du}{u}$$

$$\int \frac{u+1}{u(1-u)} du$$

$$\int \frac{1}{u} du + 2 \int \frac{1}{1-u} du$$

$$\ln |u| + 2 \int \frac{-dw}{w}$$

$$-2 \ln |w|$$

$$w = 1-u$$

$$dw = -du$$

$$\frac{A}{u} + \frac{B}{1-u}$$

$$u+1 = A(1-u) + Bu$$

$$A=1 \quad B=2$$

$$\int \left(\frac{1}{u} + \frac{2}{1-u} \right) du$$

$$\int \frac{1}{u} du + 2 \int \frac{1}{1-u} du$$

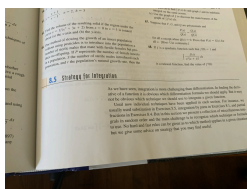
$$-2 \ln |1-u|$$

$$\ln |u| - 2 \ln |1-u| + C$$

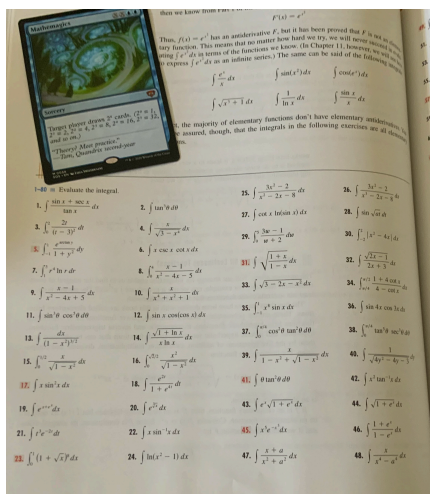
$$\ln |e^x| - 2 \ln |1-e^x| + C$$

$$x - 2 \ln |1-e^x| + C$$

part 3-



from section 8.5 of Stewart Calculus... lol
I play Se DuD and Se calculus



Pick 3

$$42 \int x^2 \tan^{-1} x dx$$

I am picking 42 mostly because I like the number 42. Given the problem has $\int x^2 \tan^{-1} x dx$

It mostly reminds me of the structure from the previous table of integration formulas, num. 13

However, because of the $\tan^{-1}$ I doubt that it's going to be that easy.

$$u = \tan^{-1} x \quad dv = x^2 dx$$

$$du = \frac{1}{1+x^2} dx \quad v = \frac{x^3}{3}$$

$$uv - \int v du$$

It's late and I just looked this one up

$$\frac{1}{3} x^3 \tan^{-1} x - \frac{1}{6} x^2 + \frac{1}{6} \ln(1+x^2) + C$$

yuck

$$33 \int \sqrt{3-2x-x^2} dx$$

I am picking 33 because it speaks to me as a historically complicated algebraic expression that I always really hated doing.

$$\int (-x^2 - 2x - 3)^{\frac{1}{2}} dx$$

no real roots, gross

$$x = -1 \pm i\sqrt{2}$$

$$7 \int_1^3 r^4 \ln r dr$$

I am picking 7 because I can already tell that there's gonna be some bs. Higher order polynomials already suck.

$$\int x^n dx = \frac{x^{n+1}}{n+1} \quad (n \neq -1)$$

$$\frac{r^5}{5}$$

$$\int_1^3 \frac{r^5}{5} \ln r dr$$